

GACHON UNIVERSITY · WEB PROGRAMMING · TEAM PROJECT

Yggdrasil

Yggdrasil — “the world tree.” A journaling website for Korean students. Write three tags — **#topic**, **@person**, **&object** — and an auto-clustering graph grows into a tree of your own life.

TEAM

Five Guys

PRESENTER · PM

Doohyun Kang

DATE

June 3, 2026

Tag **three** kinds of words → see yourself as a pattern.

Out of every word in a journal, we separate just three. Type # @ & and the word becomes a color-coded badge — structured data that powers a graph.

Topic

state · emotion · context

#exam

#travel

@ Person

the name of a person

@mom

@Minji

& Object

things · events · content

&coffee

&book

SAME JOURNAL ENTRY, WRITTEN WITH BADGES

#school I checked the exam scope. @mom
told me to eat. I read a &book .

THE PAYOFF → AUTO-CLUSTERING
GRAPH

Notes that share tags pull together **automatically**. No manual linking — recurring themes, people, and objects emerge as clusters you can click into.

A working full-stack app — not a mockup.



Auth — Signup & Login

JWT issued on login, stored client-side, guarded API routes



Folder Management

Create / read / update / delete · move notes between folders



Interactive Graph View

3D force-directed graph (three.js) · click node → open note



Most-Used Tags Panel

Top tags per type (# @ &) in the right sidebar



Note CRUD + Multi-tab

Create / list / detail / edit / delete · open notes in tabs



Tiptap Badge Editor

@ & become live badges · autocomplete suggests tags



Global & Tag Search

Search across notes · filter by tag type · ranked results



Calendar + Save Status

Monthly calendar of entries · live saved / saving status

- All wired front-to-back over a real REST API + MongoDB. Live demo next.

Five people, one clear owner per area.

MEMBER	ROLE	OWNS
Doohyun Kang	PM	Planning · schedule · documentation · conventions · decisions
Jinseo Kim	Frontend	4-pane layout · Tiptap editor · multi-tab · graph view
Yunseok Choi	Frontend	Auth UI · right panel (search · tags · calendar) · design system
Inhyeon Kim	Backend	Auth · note/folder CRUD · tag parser · search · graph API
Yushin Kim	Infra	MongoDB Atlas · Vercel · Railway · CI → frontend support

- Managed with a RACI matrix — exactly one Accountable owner per work area.

Org-level multi-repo with a full doc package.

3 REPOSITORIES · ORG GC-FIVE-GUYS

- **FiveGuys-Frontend** React + Vite client
- **FiveGuys-Backend** Express + MongoDB API
- **Docs** Documentation package

BRANCH FLOW

feature/* → **dev** → **main** · CI on every PR

📖 Docs/ — versioned document package

- **01_기획** Project plan
- **02_일정** WBS · Roles · Schedule
- **03_설계** Architecture · API · DB
- **04_개발가이드** Git rules · Conventions
- **99_기존자료** Source material
- **README.md** Reading guide

SEPARATION

3 repos: frontend · backend · docs

PROCESS

Issue-driven branches · PR review gate

DOCS AS CODE

13-doc package, versioned in Git